Atilla Saadat

Los Angeles, CA, USA

U.S. Permanent Resident (EB-1A, Extraordinary Ability)

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ResearchGate Google Scholar Github Devpost

Industry Experience

Spacecraft Guidance, Navigation, & Control (GNC) Engineer II — **Nov 2025 – Present**

Varda Space Industries — *Los Angeles, CA*

- Led Monte Carlo analysis, trajectory design optimization, and on-orbit GNC operations for W-Series 4, and post-flight reconstruction for the successful launch and reentry of W-Series 5
- Serve as GNC Responsible Engineer for Bus Block 2 (W-Series 8 and W-Series 9), architecting next-generation reentry vehicle GNC capabilities for multi-vehicle operations, mission-agnostic FAA flight safety analysis, increased mission cadence, and greater flight autonomy

Spacecraft Guidance, Navigation, & Control (GNC) Engineer — **Oct 2022 – Nov 2025**

Turion Space Corp. — *Irvine, CA*

- Led GNC engineering for Droid.001 and Droid.002 from concept through launch, commissioning, and on-orbit operations, following deployment on SpaceX Transporter-8 and Transporter-13. Developed verification and validation (V&V) methods spanning pointing budgets, power generation analysis, on-orbit diagnostics, and flight-data validation, achieving $<0.05^\circ$ (2σ) pointing accuracy on Droid.001 for Space Situational Awareness (SSA) and $<6''$ (2σ) on Droid.002 for Non-Earth Imaging (NEI).
- Designed and implemented a Python-based autonomous NEI target tracking and mission-planning algorithm for inter-satellite imaging, optimizing attitude profiles while enforcing Sun/Earth exclusion constraints and enabling **1,022 successful tasking collects and 26,467 high-precision images** across Droid.001 and Droid.002.
- Built high-fidelity spacecraft simulations with Hardware/Software-in-the-Loop (HITL/SITL) capabilities to validate autonomous inter-satellite imaging operations, evaluate mission design trades, and prototype new concepts of operations. Conducted Monte Carlo success analyses and calibrated body-frame imagers in orbit using star field imagery to ensure sub-arcminute NEI pointing alignment.
- Implemented batch least-squares orbit determination from on-orbit multiplicative extended Kalman filter (MEKF) state estimates, uploading refined ephemeris to Space-Track for conjunction analysis and maneuver planning. Verified LEO ADCS torque/momentum authority and executed power-compliant pointing campaigns, advancing GNC autonomy across Turion's SSA and NEI mission portfolio.

Hardware Functional Safety Engineering Intern — **May 2022 – Sept 2022**

NVIDIA Corp. — *Santa Clara, CA*

- Automated Fault Tree Analysis (FTA) across over 100,000 failure modes in Python, deployed in production for the Mercedes-Benz × NVIDIA Jetson AGX Orin system-on-chip (SoC) under ISO 26262 (Road Vehicles – Functional Safety)
- Streamlined failure analysis workflows for autonomous vehicle development, improving the traceability and repeatability of functional safety evidence across hardware fault campaigns

Robotic Controls Engineering Intern II — **Sept 2019 – Jan 2020**

Mujin Inc. — *Tokyo, Japan*

- Developed a torque model validation algorithm with tailored metrics and fault detection, improving torque coefficient estimate confidence by over 70%
- Extended and productionized the dynamics identification technique for the Mujin Controller, optimizing torque model coefficient calculations for production robots across Mujin's industrial fleet
- Collaborated with Mujin's legal department to file two proprietary software patent applications, active in the USA and Japan and pending in China

Robotic Controls Engineering Intern I — **May 2017 – May 2018**

Mujin Inc. — *Tokyo, Japan*

- Designed a dynamics identification technique for the Mujin Controller, computing friction, center of mass, and inertia tensor coefficients to enhance robot torque modeling accuracy
- Developed and deployed production-grade code validated against experimental results, improving robot trajectory tracking accuracy under high-acceleration scenarios by over 85%
- Integrated robot armature trajectory generation with data analytics and custom UX/UI visualizations, streamlining operator workflows and improving system observability

Space Systems Engineering Intern (Lunar Exploration) — **May 2016 – Sept 2016**

Canadensys Aerospace Corp. — *Toronto, Ontario*

- Led development of a remotely-operated lunar rover prototype integrating stereo, fisheye, and 3D camera modalities on a $10k budget, supporting lunar surface exploration objectives
- Designed and integrated a ground station GUI combining real-time camera feed visualization with vehicle command and control, improving operational efficiency for remote rover operations

Education

M.S. – Computer Science — **Jan 2025 – Apr**

Georgia Institute of Technology — *Atlanta, Georgia*

- Specialization: Artificial Intelligence; coursework and projects emphasize deep learning, reinforcement learning, robotics AI, and space computing for autonomous aerospace systems

M.A.Sc. – Aerospace Science & Engineering — **Sept 2020 – Aug 2022**

University of Toronto — *Toronto, Ontario*

- Spacecraft Systems Engineering and Attitude Determination & Control Systems (ADCS) research and development at the UTIAS Space Flight Laboratory (SFL)
- Thesis: Design and Test Optimizations for Spacecraft Attitude Determination and Control Subsystems
- Research Advisor: Dr. Robert E. Zee
- Awarded Research Fellowship & Internal Student Fellowship. Graduated with "High Distinction"

B.A.Sc. – Honours Mechanical Engineering w/ Aerospace Option — **Sept 2015 – Aug 2020**

University of Windsor — *Windsor, Ontario*

- Aerospace ML Researcher at the Intelligent Control, Analysis, and Modeling (iCAM) Lab
- Research Advisor: Dr. Afshin Rahimi
- Dean's List (3x)

Technical Skills

Research Areas: Spacecraft GNC, ADCS, state estimation, numerical optimization, trajectory optimization, orbit determination, autonomous mission planning, embedded spacecraft systems, CubeSats, distributed space systems, robotics, machine learning

Languages: Python, MATLAB/Simulink, C/C++, Java, JavaScript, PHP, HTML/CSS, NASA JPL F Prime

Engineering Tools: AGI STK, FreeFlyer, GMAT, Basilisk, OpenC3 COSMOS, SolidEdge, Siemens NX w/ Space Systems Thermal, SolidWorks, Catia V5, KiCad, Altium

ML/Scientific Computing: NumPy, SciPy, Matplotlib, Plotly, Scikit-learn, OpenCV, TensorFlow, Keras, TSfresh, Hyperopt, pandas, PyTorch

Developer Tools: Linux, Git, Bash, Docker, PyTest, Google Cloud Platform, AWS, MongoDB, MySQL, PostgreSQL, VS Code, Eclipse, Android Studio, LaTeX

Fabrication and Lab: 3D printing, MIG/TIG/Stick welding, wire harnessing, multimeters, oscilloscopes, ESD compliance, SMT soldering, Raspberry Pi/Arduino/BeagleBone

Research Experience

Graduate Researcher — Sept 2020 – Sept 2022

UTIAS Space Flight Laboratory — *University of Toronto*

- Automated ADCS hardware testing (rate sensors, magnetometers, magnetorquers, fine sun sensors, reaction wheels) across 16 satellites in the HawkEye 360 constellation, reducing test time by 63% through Python software and hardware workflow redesign, producing repeatable V&V methods, improved data quality, and stronger flight-readiness evidence
- Developed a novel Earth horizon sensor using a repurposed spacecraft inspection camera, achieving 1° RMSE nadir-vector estimation accuracy via Python/OpenCV image processing and validating the method against STK EOIR simulation and on-orbit NorSat-2 imagery. Presented at the 2022 Small Satellite Conference, winning **1st place** in the student competition
- Developed an in-situ inertia tensor estimation method for on-orbit satellites, validated through MATLAB/Simulink ADCS simulation, and implemented new reaction-wheel attitude commands in C++ flight code for the HawkEye 360 mission. Related conference paper selected by the Canadian Space Agency (CSA) as one of two Canadian delegate submissions to the IAC 2022 E2 Student Conference
- Built a model-based solar power generation analysis tool in Python and STK for NASA MSFC's StarBurst mission, incorporating photovoltaic self-shadowing effects from the SolidEdge CAD model. Automated analysis across 135+ attitude and orbital cases with shadowed string thresholding, producing high-fidelity power budget reports

Undergraduate Researcher — Mar 2019 – Oct 2020

Intelligent Control, Analysis, and Modeling (iCAM) Laboratory — *University of Windsor*

- Developed a novel ensemble-based ML algorithm for fault detection in satellite reaction wheels, using scikit-learn, TensorFlow, and Keras for fault case classification and feature extraction
- Conducted time-series data analysis research, optimizing dataset generation and establishing best practices for feature extraction in ML-based fault diagnosis applications
- Co-authored published papers and conference presentations advancing the field of intelligent control and fault diagnosis for spacecraft systems

Awards

4th Place in MIT STORM-AI Challenge | *Massachusetts Institute of Technology (MIT) ARCLab* — **Jul 2025**

- Built a metric-aligned BiGRU-Attention model for real-time 72-hour thermospheric density forecasts, exceeding transformer baselines at millisecond latency and demonstrating ML-enabled autonomy for space situational awareness and onboard planning

Payload Pioneer 30 under 30 Award | *Payload Space* — **Oct 2024**

- Recognized by Payload's "30 under 30" for achievements across the space industry, including innovation, advocacy, and leadership

SSPI 20 under 35 Award | *Space & Satellite Professionals International (SSPI)* — **Sept 2024**

- Recognized by SSPI's "20 under 35" for outstanding contributions and leadership in the space industry

1st place in Frank J. Redd Student Competition | *Small Satellite Conference* — **Aug 2022**

- Awarded $10k for best paper and presentation in an international student competition

Canadian Delegate in IAC Student Conference | *Canadian Space Agency (CSA)* — **Aug 2022**

- Selected as one of two students across Canada to attend the IAC 2022 50th Student Conference by the CSA

Research Fellowship & Internal Student Fellowship | *University of Toronto* — **Sept 2020**

- Awarded fully-funded tuition and stipend for admittance into UTIAS Space Flight Laboratory

1st place team in CSDC Critical Design Review | *Canadian Satellite Design Challenge (CSDC)* — **Oct 2019**

- Presented CDR-level designs for a 3U Cube Satellite to CSA, MDA, and ABB spacecraft engineers

Ontario Aerospace Council Scholarship | *Ontario Aerospace Council* — **May 2019**

- Awarded one of four awards to Ontario aerospace engineering students for academic and extra-curricular achievements related to aerospace studies

Dean of Engineering Conference Grant | *University of Windsor* — **Jan 2019**

- Awarded directly by the Dean of Engineering to attend the ESA Atlantic from Space Workshop

ESA CryoSat-2 Science Meeting Student Award | *Canadian Space Agency (CSA)* **Mar 2017**

- Awarded to select students across Canada to attend ESA CryoSat-2 meeting as student representatives for the CSA, used for research and resources for CryoRoute

P.E.O. Windsor-Essex Chapter Bursary | *Professional Engineers of Ontario (P.E.O)* **Mar 2016**

- Awarded to 1st year engineering students who have demonstrated excellent academic achievement

University of Windsor Scholarships & Bursaries

$16k total – Awarded for undergraduate academic achievement and extra-curricular activities, unless otherwise stated:

- Student Access Bursary | Jun 2020 & Jun 2019
- Ontario First Generation Bursary | Jun 2019 & Sept 2018
- UWindsor In-Course Bursary | Jun 2019
- Student Life Enhancement Fund | Jan 2019 | Awarded to attend ESA Atlantic from Space Workshop
- Helen Norma Laframboise Scholarship | Sept 2018
- Faculty Retirees' Scholarship | Sept 2018
- Ontario Student Opportunity Trust Fund (OSOTF) Scholarship | Sept 2018
- Marian McLean Campus Spirit Scholarship | Sept 2018 | Awarded for WinSAT events held on campus
- International COOP Bursary | Jan 2018 | Awarded for Mujin Inc. internship in Japan
- Duronio Family Scholarship | Jan 2017
- University of Windsor Entrance Scholarship | Sept 2015
- Golden Key International Honour Society | Sept 2016 | Nomination

Master's Thesis

Saadat, A. (2022). *Design and Test Optimizations for Spacecraft Attitude Determination and Control Subsystems.* Master's thesis, University of Toronto Institute for Aerospace Studies, Space Flight Laboratory. doi:10.1807/125667

Peer-Reviewed Journal Articles

Rahimi, A., & **Saadat, A.** (March, 2020). Fault isolation of reaction wheels onboard three-axis controlled in-orbit satellite using ensemble machine learning. *Aerospace Systems*, 3(2), 119–126. doi.org/10.1007/s42401-020-00046-x

Conference Proceedings & Presentations

Saadat, A., V. Bhosale, K. Bhardwaj, A. Gavrilovska (January, 2027). An Adaptive Telemetry Probe for Resource-Constrained Spacecraft Using F Prime. *AIAA SciTech Forum 2027*, Orlando, Florida. (*Accepted*)

Saadat, A., V. Chen, J. Austin (September, 2025). End-to-End Autonomous Mission Planning and Spacecraft Attitude Optimization for Resident Space Object Imaging. *Advanced Maui Optical and Space Surveillance Technologies (AMOS) Conference 2025*, Maui, Hawaii. [PDF]

Saadat, A. Attitude Determination with Self-Inspection Cameras Repurposed as Earth Horizon Sensors. *Frank J. Redd Student Competition - Small Satellite Conference 2022*, Utah State University, Logan, Utah. (1st Place Paper and Presentation) [PDF]

Saadat, A. On-orbit Spacecraft Inertia Tensor Estimation. *International Astronautical Congress 2022*, Paris, France. [PDF]

Folami, F.O., Rahimi, A. & **Saadat, A.** (March, 2020). Fault Isolation of Reaction Wheels on-board 3-axis Controlled in-orbit Satellite Using Machine Learning Techniques. IAC-20.B4.9-GTS.5.10, *International Astronautical Congress 2020*, Dubai (Online), United Arab Emirates. [PDF]

Rahimi, A., & **Saadat, A.** (July, 2019). Fault isolation of reaction wheels on-board 3-axis controlled in-orbit satellite using ensemble machine learning techniques. *The International Conference on Aerospace System Science and Engineering*, Toronto, Ontario. [PDF]

Technical Reports

Saadat, A., Weeratunga, M., Hasan, M., & Mehroke, A. Reconstructing ESA Time-Series Trojans via Fourier-Regularized Deep Learning. Technical Report, CS7643 Deep Learning Final Report, Georgia Institute of Technology, July 2025. DOI: 10.13140/RG.2.2.17228.68480. [PDF]

Saadat, A. Real-Time Thermospheric Density Forecasting with a Metric-Aligned BiGRU-Attention Network for Space Situational Awareness. Technical Report, MIT ARCLab AI Competition, May 2025. [PDF]

Patents

Saadat, A., & Usui, K. (May, 2020). Method And Computing System For Estimating Parameter For Robot Operation. Patent applications in USA (Active), Japan (Active), and China (Pending)

Saadat, A., & Usui, K. (May, 2020). Method And Computing System For Determining A Value Of An Error Parameter Indicative Of Quality Of Robot Calibration. Patent applications in USA (Active), Japan (Active), and China (Pending)

Poster Presentations

Saadat, A., V. Chen, J. Austin (September, 2025). End-to-End Autonomous Mission Planning and Spacecraft Attitude Optimization for Resident Space Object Imaging. *Advanced Maui Optical and Space Surveillance Technologies (AMOS) Conference 2025*, Maui, HI

Saadat, A. (March, 2019). CryoRoute - Generating Globally-Optimized Naval Routes in the Arctic Ocean. *UWillDiscover Conference*, University of Windsor, Windsor, ON

Saadat, A. (January, 2019). CryoRoute - Generating Globally-Optimized Naval Routes in the Arctic Ocean. *ESA's Atlantic from Space Workshop*, National Oceanography Centre, Southampton, UK

Academic Reviews

Judge for AIAA Region VI Student Conference 2025

Completed Graduate-level Courses

- Microsatellite Design I
- Microsatellite Design II
- Spacecraft Dynamics & Control
- Computational Optimization
- Control Theory
- Flight Dynamics & Control
- Aerospace Structures
- Aerodynamics & Performance
- Aerospace Propulsion
- Aerospace Materials & Mfg.
- CS-7643 Deep Learning
- CS-7638 Robotics: AI Techniques
- CS-7641 Machine Learning
- CS-7642 Reinforcement Learning
- CS-8903 Space Computing

Co-Curricular Experience and Projects

Real-Time Spacecraft Thermospheric Density Forecasting — **May 2025**

MIT Astrodynamics, Space, Robotics, and Controls Laboratory (ARCLab)

- Developed and open-sourced a PyTorch BiGRU-Attention model that converts 60 days of solar-wind and orbital data into 72-hour, 10-minute-step thermospheric density forecasts in milliseconds, training in $\leq$ 4h on a single RTX 3090 Ti
- Ranked **4th out of 159 participants** (0.709 normalized skill score, exceeding the transformer baseline) and achieved the highest paper score in the MIT ARCLab STORM-AI Competition, delivering a deployable real-time solution for drag prediction and collision-risk screening

President / Founder / Lead Space Systems Engineer — **2018 – 2020**

University of Windsor Space & Aeronautics Team (WinSAT)

- Founded WinSAT and led the team to **1st place** in the Canadian Satellite Design Challenge (CSDC) Critical Design Review, directing design and development of a 3U CubeSat for LEO while managing 20+ students and securing over $40k in funding
- Led end-to-end development of all satellite subsystems including Attitude Determination & Control, RF Communications, and Structural & Thermal, driving cross-disciplinary integration across a multidisciplinary team

- Organized and led educational workshops on satellite engineering and Amateur Radio Operator certification for 30+ students, expanding hands-on technical capacity within the university community

Lead Avionics Engineering - Sounding Rocket **2015 – 2019**

Rocketry Payload Division, UWindsor Rocketry Team

- Designed and built a custom aerodynamics data acquisition module using Raspberry Pi and a multi-sensor array, enabling precise in-flight aerodynamic analysis and performance validation
- Integrated custom PCB design with embedded software and redundancy handling to reliably capture and process high-rate rocket telemetry including acceleration, angular velocity, and atmospheric conditions

CryoRoute - Globally Optimized Naval Routes in the Arctic Ocean **Jan 2019**

Independent Research Project

- Developed a novel route optimization tool for Arctic naval navigation, using CryoSat-2 SAR sea ice freeboard data from the Centre for Polar Observation and Modelling (CPOM) to improve safety and transit efficiency
- Implemented a tailored A* pathfinding algorithm minimizing sea ice thickness, distance, and transit duration to produce optimized Arctic shipping routes
- Presented findings at NASA Space Apps Waterloo, UWillDiscover Conference, and ESA's Atlantic from Space Workshop

Inertial Electrostatic Confinement (IEC) Nuclear Fusion Reactor Demonstrator **Aug 2015**

Independent Research Project

- Designed and built a vacuum chamber to simulate atmospheric gas ionization and plasma generation, replicating core processes of a deuterium-powered IEC nuclear fusion reaction
- Operated a 15 kV high-voltage power supply under strict safety protocols to maintain stable plasma conditions throughout experimentation
- Characterized the relationship between internal grid geometry and plasma formation rates using visual luminescence techniques, laying groundwork for future neutron radiation simulations and fusion validation

Professional Training & Certification

Introductory and Advanced F Prime Workshop Certification **Feb 2026**

NASA Jet Propulsion Laboratory *Pasadena, CA*

Nationally Registered Emergency Medical Technician **Sept 2025**

National Registry of Emergency Medical Technicians (NREMT) *United States*

PADI Rescue Diver w/ Emergency First Response (EFR) **Oct 2024**

Professional Association of Diving Instructors (PADI) *Laguna Beach, CA*

Basic Life Support (CPR & AED) **Sept 2024**

American Heart Association *United States*

PADI Advanced Open Water Diver **Apr 2024**

Professional Association of Diving Instructors (PADI) *Laguna Beach, CA*

- Specialties: Deep Dive, Search & Recovery, Peak Performance Buoyancy, U/W Navigation, U/W Naturalist

PADI Enriched Air Nitrox Diver **Apr 2024**

Professional Association of Diving Instructors (PADI) *Laguna Beach, CA*

Wilderness Medicine - First Aid / Epinephrine Auto-injector **Apr 2024**

NOLS Wilderness Medicine *Los Angeles, CA*

PADI Open Water Diver **Feb 2024**

Professional Association of Diving Instructors (PADI) *Laguna Beach, CA*

Canadian Amateur Radio Operator Certificate - Callsign: VA3UWS **Jan 2019**

Innovation, Science and Economic Development Canada *Government of Canada*

- Basic with Honours - Access to all amateur radio frequency bands

Siemens NX Space Systems Thermal Training **Feb 2019**

CSDC Structural & Thermal Analysis Workshop *Magellan Aerospace, Winnipeg, Manitoba*

- Learned spacecraft structural and thermal analysis techniques for small satellites in Siemens NX

Hackathon Projects & Awards

GeoRisk **2017**
Winter Hackathon *University of Windsor, Windsor, Ontario*

- Developed a Google Maps web-based application to analyze client data for a sponsoring insurance company
- **Awards: UWindsor EpiCentre Startup Series-A Funding ($6k), later used to seed WinSAT**

Lubdub **2017**
SpartaHacks III *Michigan State University, East Lansing, MI*

- Developed a web application that records and detects heart defects using audio wave analysis and a stethoscope, achieving real-time diagnostic capabilities
- **Awards: SpartaHacks 2nd Place Winner, ThoughtWorks Honorable Mention**

Gateway 2.0 **2015**
Facebook Global Hackathon *Facebook HQ, Menlo Park, CA*

- Extended an iOS messaging app with embedded API search query responses and natural language processing

Gateway **2015**
MHacks 6 *University of Michigan, Ann Arbor, MI*

- Developed an iOS messaging app with embedded API search query responses and natural language processing
- **Awards: Best Use of Facebook Parse API (Invitation to Facebook Global Hackathon), Best iOS App (sponsored by Apple Inc.), Best Use of Expedia API**

Professional Affiliations & Community Engagement

Aerospace Education Officer, 2d Lt **Apr 2024 – Present**
Civil Air Patrol (CAP) - Squadron 232 *Mission Viejo, CA*

- Deliver STEM education through lectures, hands-on activities, and rocketry/Arduino projects to inspire cadets in aerospace and engineering fundamentals.

Programs Chair **June 2024 – Present**
American Institute of Aeronautics and Astronautics (AIAA) - Orange County Chapter *Orange County, CA*

- Coordinate and host distinguished guest speakers to advance aerospace dialogue and professional development within the local AIAA community.

Media Coverage

STARFIRE: FLEET COMMAND | *Turion Space Corp.* **Sept 2024**
Droid.001 Alpha Centauri Imaging | *Turion Space Corp.* **Jun 2024**
Droid.001 Moon Image Results | *Turion Space Corp.* **Apr 2024**
Droid.001 Mission and Simulation Highlights | *Turion Space Corp.* **Sept 2023**
Small Satellite Student Competition Award | *University of Toronto Engineering Dept.* **Aug 2022**
UWindsor satellite design leading student competition | *University of Windsor Engineering Dept.* **Nov 2019**
WinSAT wins critical satellite design review | *Windsor Star – News Outlet* **Nov 2019**
Aerospace engineering student aiming high | *University of Windsor Engineering Dept.* **Aug 2019**
Mujin Inc. Internship - Student Spotlight | *University of Windsor Co-operative Education Dept.* **Oct 2018**